

THE STORY OF OCEANOGRAPHY

ONE OF THE WORLD'S MOST UNEXPLORED REALMS HAS GRADUALLY BEEN REVEALING ITS SECRETS

For a long time, oceans have been the least understood features on Earth. However, knowledge of marine life and the topography of the ocean floor has gradually accumulated and, in recent years, new exploration techniques have led to several discoveries.

The earliest records of sea exploration date back 3,000 years to the Phoenicians, who made charts to navigate and used weights to plumb the oceans' depths. Ancient Greek philosopher Aristotle was one of the first to speculate about marine life, and other Greeks developed instruments to help ships locate their position when far from shore.

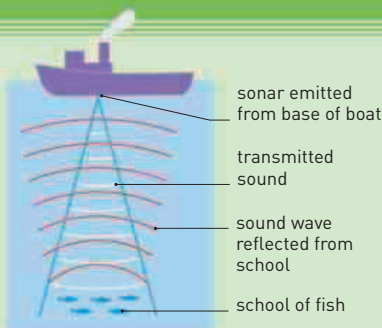
However, the open oceans remained unexplored by Westerners until the 1400s, when Christopher Columbus sailed westward into the Atlantic in the hope of finding land on the far side. This paved the way for further voyages of exploration, such as

Ferdinand Magellan's circumnavigation, which finally revealed the extent of the world's oceans and allowed mapmakers to chart their shapes.

Scientific attempts to study beneath the surface of the oceans began in the 19th century. Early surveys were conducted using sounding chains and sample nets. The introduction of sonar after World War II helped in mapping the ocean floor. Recently, improved sonar, satellite techniques, and an array of submersibles have helped add to our knowledge of marine life and ocean currents as well as the oceans' geography.

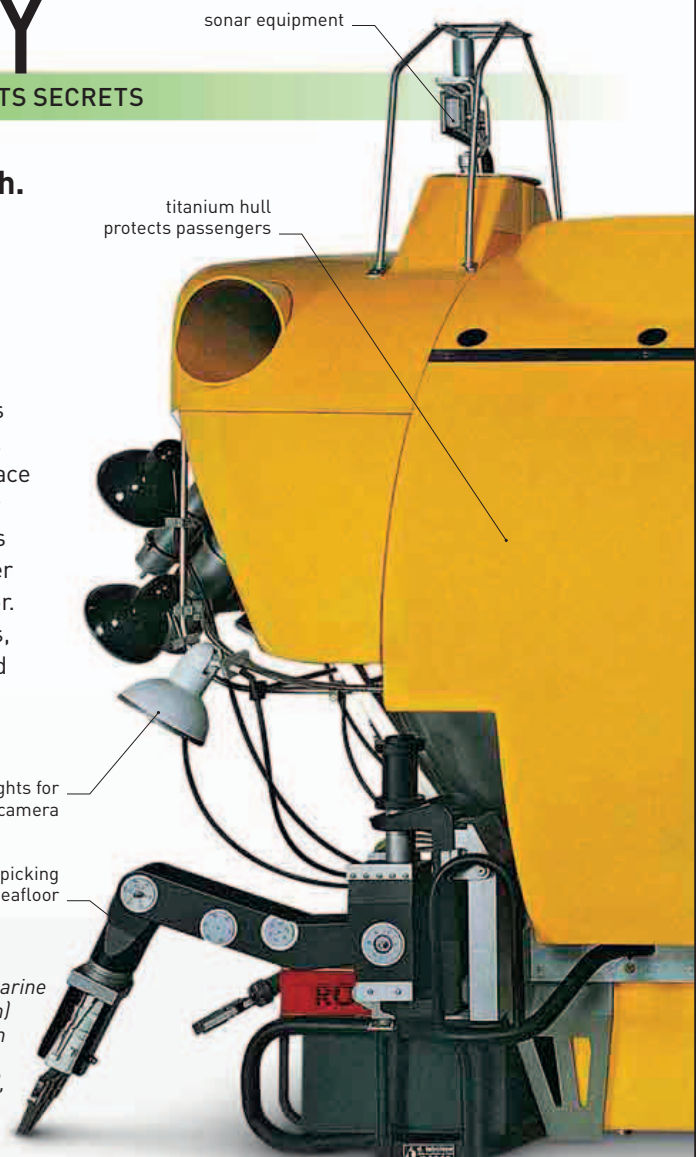
SONAR

The first maps of the ocean floor were produced by sonar, a technology developed during World War II to detect submarines by picking up reflections of sound waves from underwater objects. It can also be used to detect schools of fish. Nowadays, new systems, such as side-scan sonar, are being used together with GPS to survey vast areas quickly.



Nautile

French miniature submarine Nautile is only 26 ft (8m) long. However, its tough hull allows it to reach a depth of 3¾ miles (6km), and external robotic arms, video cameras, and floodlights enable detailed exploration.



1200-250 BCE

Phoenician traders
The first seafarers, the Phoenicians plumb the seabed to find channels. They develop the first coins to facilitate trade.



Ancient Phoenician coin

c. 80 BCE

Antikythera mechanism
The Greeks develop instruments, such as the clockwork Antikythera mechanism, to plot the movement of the heavens and to navigate at sea.

1519-22

Strait of Magellan
Portuguese explorer Ferdinand Magellan is the first to sail from the Atlantic to the Pacific Ocean, and discovers the Strait of Magellan on the way.



Map of the Strait of Magellan

1842

Matthew Maury
Considered to be the "father of oceanography," US naval officer Maury compiles sea charts of the world's oceans.

500-200 BCE

Greek marine science
Aristotle identifies many marine species, such as crustaceans, mollusks, echinoderms, and fish.



Ancient Grecian bowl showing a sailing boat

1492

Columbus' voyage
Italian navigator Christopher Columbus's voyage to the Americas shows that it is possible to cross the Atlantic and even sail around the world.



Christopher Columbus

1769-71

Captain Cook's Endeavour
British naval captain James Cook makes voyages to the Southern Oceans and is the first European to reach New Zealand and Australia.



Endeavour